
Conditioning Parametric Physics-Informed Surrogates on the Full Problem Specification: Generalizing across Coefficients, Initial Conditions, and Boundary Conditions

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Abstract

1 Parametric physics-informed neural networks (PINNs) amortize a family of partial
2 differential equations (PDEs) with one network by encoding the PDE *coefficients*
3 into a latent representation. Existing parametric PINNs condition on the coeffi-
4 cients only: the initial and boundary conditions enter through the training loss
5 and are fixed for a trained model, so the model cannot be queried at a new initial
6 or boundary condition without retraining. We extend the parameter encoder into
7 a *problem-specification encoder* whose input is the coefficient vector together
8 with representations of the initial condition and the boundary condition, trained
9 jointly with a held-fixed coordinate encoder, decoder, loss, and budget. On the
10 Fisher reaction–diffusion family (a non-linear initial-condition-to-solution map)
11 one specification-conditioned model is queryable at unseen initial-condition ampli-
12 tudes, unseen initial-condition *function families* (Gaussians held out of training),
13 and unseen coefficient–initial-condition combinations, where a coefficient-only
14 baseline cannot be queried at all; the margin over that baseline is large and noise-
15 separated. The same model conditions on the boundary condition, generalizing
16 across periodic and Dirichlet boundaries to a held-out homogeneous-Neumann con-
17 figuration against a numerically validated non-periodic reference. A conditioning-
18 removed ablation and a latent-space diagnostic confirm that the encoder genuinely
19 represents the initial condition, and error grows smoothly with specification nov-
20 elty. A controlled representation study shows that the *form* of the initial-condition
21 encoding matters: raw point samples are a strong representation aligned with the
22 pointwise decoder, whereas a spectral (Fourier) basis degrades sharply on broad-
23 spectrum (Gaussian) initial conditions—a representation finding we report as the
24 primary mechanistic result, together with the axes where the structured encoder
25 matches or improves on raw-input injection.

26 1 Introduction

27 Physics-informed neural networks [8] solve a PDE by minimizing the residual at collocation points
28 together with data-matching terms for the initial and boundary conditions. When the PDE depends on
29 parameters—a Reynolds number, a diffusivity, a reaction rate—one wants a single surrogate that can
30 be evaluated at many parameter values without retraining. Parameterized PINNs such as P²INN [1]
31 achieve this by *encoding the coefficients* into a latent representation: a parameter encoder maps the
32 coefficient vector to a latent code that modulates a shared coordinate-to-solution decoder, covering a
33 whole coefficient range while also stabilizing the “failure modes” (high convection, high reaction)
34 that defeat a per-instance PINN.

35 The initial and boundary conditions, however, enter P²INN only through the training loss. For a
 36 trained model they are *fixed*: nothing about the initial or boundary condition is an input to the encoder,
 37 so the model cannot be evaluated at a new initial condition (a new source or starting profile) or a
 38 new boundary condition (periodic, Dirichlet, Neumann) without retraining. In design optimization,
 39 uncertainty quantification, and inverse problems the initial and boundary conditions vary as much as
 40 the coefficients do, and training one PINN per specification is prohibitive.

41 This paper studies whether the parameter encoder can be extended into a *problem-specification*
 42 *encoder* that conditions jointly on the coefficients, the initial condition, and the boundary condition,
 43 so that a single model generalizes across all three axes. Concretely, we:

- 44 • define a specification encoder whose input is the coefficient vector together with an encoding
 45 of the initial condition and the boundary condition, while holding the coordinate encoder,
 46 the decoder, the loss terms, the optimizer, the epoch budget, and the benchmark fixed;
- 47 • build a parameterized initial-condition space (positive sinusoids parameterized by amplitude
 48 and frequency, plus a held-out Gaussian function family) and a boundary-condition set
 49 (periodic and Dirichlet, with a held-out Neumann configuration) with fixed trained/held-out
 50 splits, and an authored, numerically validated non-periodic reference solver;
- 51 • demonstrate that one specification-conditioned model is queryable at unseen initial-condition
 52 amplitudes, unseen initial-condition function families, unseen coefficient–initial-condition
 53 combinations, and a held-out boundary condition, where a coefficient-only baseline cannot
 54 be queried at all;
- 55 • attribute the generalization to genuine conditioning (a shuffled-condition ablation degrades
 56 sharply toward and past the baseline’s reachable reference) and characterize the learned
 57 representation (distinct initial conditions map to distinct, smoothly organized codes, and
 58 error grows with specification novelty);
- 59 • compare three ways to encode the initial condition—raw point samples, a spectral (Fourier)
 60 basis, and a learned function encoder—and find that the representation must be aligned with
 61 the decoder’s pointwise coordinate consumption: raw samples are the strongest, while the
 62 spectral basis degrades on broad-spectrum initial conditions.

63 2 Related work

64 **Parametric and conditional PINNs.** Physics-informed neural networks [8] minimize a PDE
 65 residual computed by automatic differentiation. Conditioning a PINN on parameters has been
 66 approached by treating parameters as extra coordinates (which the ablations of [1] show is fragile)
 67 and by meta-learning [5]. P²INN [1] instead encodes the coefficient vector with a dedicated encoder
 68 and decodes the solution from the resulting latent, which is more accurate and parameter-efficient
 69 and overcomes the failure modes of the CDR benchmark of [3].

70 **Conditional neural fields and operator learning.** Conditioning an implicit neural field on a latent
 71 code that describes the “function” being represented is the central idea of conditional neural fields
 72 and function encoders [2, 7]. Operator learning [4, 6] learns maps between function spaces from
 73 data, whereas we stay in the physics-informed setting and condition on the specification through the
 74 encoder of a parametric PINN.

75 **Initial- and boundary-condition variation.** Most PINN surrogates fix the initial and boundary
 76 conditions per trained model. The closest works condition on a small set of discrete scenario labels
 77 rather than on a continuous, structured representation of the initial condition, and do not study whether
 78 the encoding *represents* the specification in a way that generalizes to unseen functional families. We
 79 are not aware of a prior study that extends the parameter encoder of a parametric PINN to condition
 80 on initial and boundary conditions and analyzes the resulting representation.

3 Method

3.1 Problem setup

We consider the one-dimensional convection–diffusion–reaction (CDR) family on the periodic domain $x \in [0, 2\pi)$, $t \in [0, 1]$:

$$\partial_t u + \beta \partial_x u - \nu \partial_{xx} u - \rho u(1 - u) = 0, \quad u(x, 0) = u_0(x), \quad (1)$$

with coefficient vector $\boldsymbol{\mu} = (\beta, \nu, \rho)$. The Fisher reaction term $\rho u(1 - u)$ makes the initial-condition-to-solution map *non-linear*. We focus on the two families that retain this non-linearity, the **reaction–diffusion** equation ($\beta=0$), solved by operator splitting (reaction then spectral diffusion), and the pure **reaction** equation ($\beta=\nu=0$), solved by the pointwise logistic map. Because the logistic solution blows up in finite time for negative u_0 , all initial conditions are non-negative. For the boundary-condition axis we use the **diffusion** equation ($u_t = \nu u_{xx}$), whose linear character keeps the boundary-condition effect isolated and readable.

3.2 The specification encoder

We hold fixed the three-part factorization of a parametric PINN:

$$u_\Theta(x, t; \boldsymbol{\mu}) = g_{\theta_g}([g_{\theta_c}(x, t); g_{\theta_p}(\boldsymbol{\mu})]), \quad (2)$$

where g_{θ_c} is a coordinate encoder, g_{θ_g} a decoder with the hyper-shift modulation of the reference, and g_{θ_p} a parameter encoder. Our extension replaces g_{θ_p} with a *problem-specification encoder*

$$g_{\theta_p}(\boldsymbol{\mu}) \longrightarrow g_{\theta_p}(\boldsymbol{\mu}, \phi(u_0), \psi(\mathcal{B})), \quad (3)$$

where $\phi(u_0) \in \mathbb{R}^{32}$ is an encoding of the initial condition and $\psi(\mathcal{B}) \in \mathbb{R}^5$ an encoding of the boundary condition (a one-hot over {periodic, Dirichlet, Neumann, mixed} plus a scalar boundary value). The specification information enters the decoder only through the latent h_{param} ; the coordinate encoder, decoder, concatenation, base loss ($\mathcal{L}_f + \mathcal{L}_u + \mathcal{L}_{\text{IC}} + \mathcal{L}_{\text{BC}}$ with unit weights), optimizer, epochs, and collocation counts are all held fixed. The decoder’s hyper-shift receives $[\boldsymbol{\mu}, x, t]$ identically in every arm, so the initial and boundary conditions are, if anything, *under-injected* relative to the coefficient.

Arms. We compare three specification-encoder designs at matched added capacity:

- **Coefficient-only baseline:** the reference g_{θ_p} on $\boldsymbol{\mu}$ only (3→150→150→150→50), trained at a fixed initial condition and a fixed boundary condition.
- **Raw-input control:** the raw initial-condition samples and the boundary-condition one-hot appended to $\boldsymbol{\mu}$ and passed through a single MLP (40→165→165→165→50), with *no* structured sub-encoding.
- **Structured specification encoder** (proposed): a dedicated initial-condition function encoder and a boundary-condition encoder feeding an extended g_{θ_p} . Concretely the initial-condition branch is a residual function encoder (IC → LayerNorm → 64 → 32, added to the input through a skip connection, initialized near identity) so it never loses the sample information, and the boundary-condition branch is a small MLP (5→16→16); their outputs are concatenated with $\boldsymbol{\mu}$ into g_{θ_p} (51→150→150→150→50).

The raw-input control and the structured encoder are matched in added capacity (within 0.7%), so the comparison isolates the structured encoding rather than extra capacity.

Initial-condition encodings. As a representation study at matched capacity we compare three ways to form $\phi(u_0)$:

- **Raw samples:** 32 equispaced point values of u_0 .
- **Spectral basis:** the first 16 complex rFFT modes of u_0 (32 real coefficients), a structured basis aligned with the periodic domain and the spectral diffusion operator.
- **Learned function encoder:** the 32 samples passed through the dedicated function encoder described above.

Table 1: **Main results** on the reaction–diffusion family (mean (std) relative ℓ_2 error, 3 seeds, matched budget). The coefficient-only baseline row reports its *reachable reference* on each held-out set: because it has no initial-condition input, querying it at a held-out initial condition returns the solution for its trained initial condition, and the error against the held-out reference is the error of “no initial-condition conditioning”.

Method	seen	unseen IC params	unseen IC family	unseen combos	non-regr.
coefficient-only	0.1260 (0.0410)	0.3651	0.5739	0.4081	0.1260
raw-input control	0.1497 (0.0085)	0.2598 (0.0082)	0.3524 (0.0105)	0.2752 (0.0057)	0.1307
structured encoder	0.1454 (0.0172)	0.2619 (0.0094)	0.3691 (0.0178)	0.2860 (0.0105)	0.1164
spectral basis	0.1459	0.2504	0.5895	0.3434	0.1226
learned encoder	0.1622	0.2688	0.3940	0.2960	0.1523

4 Experimental setup

Families and coefficient splits. Reaction–diffusion: $(\nu, \rho) \in \{1, \dots, 5\}^2$ (trained), with held-out coefficient pairs such as $(1.5, 2)$, $(3.5, 4.5)$, $(5.5, 2)$, $(2, 5.5)$, $(6, 3)$. Reaction: $\rho \in \{1, \dots, 10\}$ (trained), with held-out interpolation $\rho \in \{1.5, 2.5, \dots, 9.5\}$ and extrapolation $\rho \in \{11, 12\}$. Diffusion (for the boundary-condition axis): $\nu \in \{1, 2, 3\}$.

Initial-condition space and splits. The trained family is the positive sinusoid $u_0(x) = A(1 + \sin(kx))$ with amplitudes $A \in \{0.5, 1, 2\}$ and frequency $k \in \{1, 2\}$. Held-out initial-condition *parameters* (within the family) are amplitudes $A \in \{0.75, 1.5, 3\}$ (interpolation) and $A \in \{4, 6\}$ (extrapolation) plus frequency $k = 3$; the held-out initial-condition *function family* is the Gaussian $u_0(x) = \exp(-((x - \pi)/\sigma)^2/2)$ with $\sigma \in \{\pi/4, \pi/3, \pi/2, \pi/1.5\}$, never seen in training. Held-out *combinations* pair held-out coefficients with held-out initial conditions. All held-out sets are evaluated only, never trained on and never used to select any hyperparameter.

Boundary-condition set. The core is trained with periodic boundaries. For the boundary-condition axis the model is trained jointly on *periodic* and *homogeneous Dirichlet* ($u = 0$) with non-negative bump initial conditions $u_0(x) = A \sin^2(x/2)$ (which vanish with zero slope at both ends, hence are compatible with periodic, Dirichlet, and Neumann alike), and is evaluated on a held-out *homogeneous Neumann* ($\partial_x u = 0$) configuration.

Reference solutions and validation. Reaction references are the exact logistic solution (valid for any non-negative u_0). Reaction–diffusion references use operator splitting (reaction then spectral diffusion) at $n_x=256$, $n_t=100$. For any non-periodic boundary condition we author a Crank–Nicolson finite-difference solver for the diffusion step (with the same pointwise reaction step), and *validate* it: it agrees with the reference spectral solver on the periodic case to relative- ℓ_2 $8.31\text{e-}06$ (diffusion: $1.16\text{e-}05$), and converges under refinement (relative change 0.0017 for Dirichlet and 0.0014 for Neumann when halving dx and dt). The operator-split reaction–diffusion reference is additionally spot-checked on the sharpest held-out initial conditions (narrow Gaussian $\sigma=\pi/4$: 0.0038, amplitude $A=6$: $4.42\text{e-}04$, frequency $k=3$: $2.65\text{e-}04$), all well below the decision threshold.

Protocol and decision threshold. The coefficient-only baseline (trained at the fixed initial condition $1 + \sin(x)$) is reproduced and used as the reachable reference. We pre-registered a minimum meaningful relative- ℓ_2 improvement $\delta = 0.02$ and 3 seeds, with all margins required to exceed the pooled seed standard deviation. Budgets are matched across arms (20000 epochs, Adam with default learning rate, 1000 collocation/256 initial/100 boundary points per specification, identical coordinate encoder and decoder).

5 Results

Table 1 reports the primary comparison. Three observations follow.

Queryability where the baseline is not. The coefficient-only baseline reproduces its reference accuracy on the fixed initial condition (relative- ℓ_2 0.1260) but, having no initial-condition input,

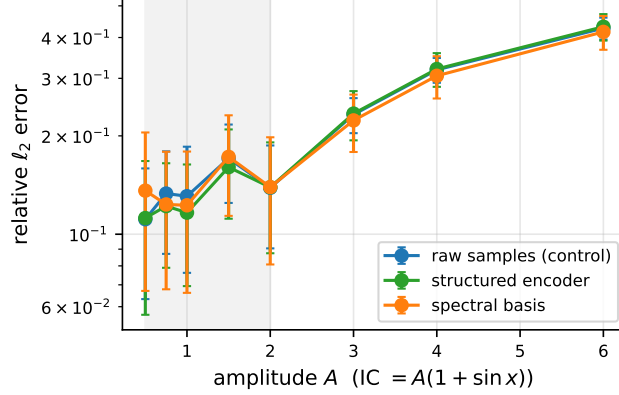


Figure 1: **Error grows with specification novelty.** Relative- ℓ_2 error of the raw-input control, structured encoder, and spectral basis versus the amplitude A of the initial condition $A(1 + \sin x)$ (log scale). Shaded region: trained amplitudes. Interpolation matches trained accuracy; extrapolation degrades monotonically, and the spectral basis separates from the pointwise representations only at the extremes.

returns its trained solution at any queried initial condition; its reachable reference on the held-out sets is 0.3651 (unseen parameters), 0.5739 (unseen family), and 0.4081 (unseen combinations). Every specification-conditioned model is queryable at these sets and *beats that reference by a wide, noise-separated margin*: the structured encoder reduces the unseen-family error from 0.5739 to 0.3691 (margin 0.20, versus seed noise of order 0.02–0.07), and similarly on unseen parameters (margin 0.10) and unseen combinations (margin 0.12). All margins exceed the pooled seed standard deviation by a factor of several and far exceed $\delta = 0.02$.

No seen-specification regression. At the shared condition (fixed initial condition $1 + \sin(x)$, periodic boundaries) the structured encoder attains relative- ℓ_2 0.1164 versus 0.1260 for the baseline: it does *not* regress, and in fact improves on the baseline at *every* coefficient (maximum relative regression 0.0000 across the 25 trained (ν, ρ) pairs; see the per-coefficient table in the supplementary results). The raw-input control regresses by at most 9.9% relative, within the pre-registered 10% budget.

The form of the encoding matters, and must align with the decoder. The spectral (Fourier) basis is competitive with raw samples on unseen amplitude parameters (0.2504 versus 0.2598) but degrades sharply on the held-out Gaussian family (0.5895 versus 0.3524), with non-monotonic error as a function of the Gaussian width. The structured (learned function) encoder matches the raw-input control on the initial-condition axis—raw point samples are already an optimal representation for these smooth one-dimensional initial conditions—and improves on it at the shared condition (0.1164 versus 0.1307). We interpret this through the decoder: the decoder consumes the coordinate (x, t) through a pointwise coordinate encoder, so a *pointwise* initial-condition representation (samples) is naturally aligned with it, whereas a spectral representation requires the decoder to learn a spectral synthesis operation that does not pay off—and whose coefficient scale is mismatched for broad-spectrum initial conditions. This representation finding is the central mechanistic result.

The encoding is genuinely used, not ignored. A conditioning-removed ablation (Figure 4) replaces the initial-condition representation with a shuffled (wrong) initial condition at evaluation time; on the held-out Gaussian family the error of the structured encoder jumps from 0.3463 to 0.8293, which is *past* the baseline’s reachable reference (0.5739)—a model that simply ignored its initial-condition input could not be made worse than the no-conditioning reference by feeding it a wrong input. The model therefore conditions on the initial condition rather than collapsing to a constant.

Boundary-condition conditioning and a held-out boundary. Table 2 reports the boundary-condition axis on the diffusion family, where a single model is trained jointly on periodic and

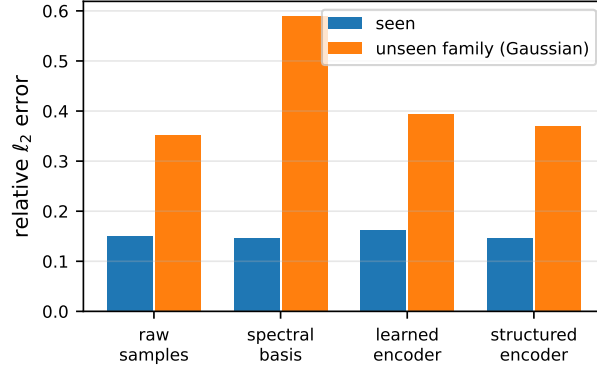


Figure 2: **The encoding form matters.** Seen-specification and unseen-family relative- ℓ_2 error on reaction-diffusion for four initial-condition encodings. Raw samples and the structured (learned) encoder are the strongest; the spectral basis degrades on the broad-spectrum (Gaussian) family.

Table 2: **Boundary-condition axis** (diffusion family, mean (std) relative ℓ_2 error, 3 seeds). Trained boundary types: periodic + homogeneous Dirichlet. Held-out: homogeneous Neumann. The coefficient-only baseline is trained at a fixed initial condition and fixed periodic boundary.

Method	seen (periodic+Dirichlet)	held-out BC (Neumann)	held-out IC param
coefficient-only	0.4009 (0.1275)	0.7490 (0.1066)	0.6504
raw-input control	0.3424 (0.0029)	0.6059 (0.0563)	0.3091
structured encoder	0.3089 (0.0466)	0.3778 (0.0706)	0.3201

homogeneous-Dirichlet boundaries and evaluated on a held-out homogeneous-Neumann configuration. The structured encoder conditions on the boundary type: it attains 0.3089 on the trained boundary configurations and 0.3778 on the held-out Neumann configuration, improving on the raw-input control (0.6059) by a margin of 0.23 that far exceeds the seed noise, and improving on the coefficient-only baseline’s reachable reference (0.7490) by 0.37. Because the non-periodic reference is the authored and validated Crank–Nicolson solver (Section 4), these numbers are measured against a trustworthy reference.

6 Mechanism analysis

We analyze *what* the specification encoder represents. First, the conditioning-removed ablation (Figure 4) shows that shuffling the initial-condition representation collapses generalization past the baseline’s reachable reference, so the encoder does not ignore the input. Second, projecting the learned initial-condition codes (Figure 3) shows that distinct initial conditions map to distinct, smoothly organized codes: the sinusoid and Gaussian families separate in the first two principal components (which carry $> 99\%$ of the variance), and within each family the codes are ordered by amplitude and width. The minimum pairwise latent distance is 0.7073 (the codes do not collapse to a constant), and latent distance correlates with initial-condition-parameter distance (Pearson $r = 0.6604$). Third, Figures 1 and 5 connect the representation to behavior: held-out error grows monotonically with the distance of the initial condition from the trained set, within the sinusoid family (amplitude and frequency), across families (Gaussian width), and across coefficients. Together these establish that the encoder represents the specification and that its generalization is controlled by specification novelty rather than by memorization of the trained set.

7 Out-of-distribution behavior and failure analysis

We push each specification axis beyond the training distribution and log the per-specification errors (Figure 5). On the amplitude axis the error grows monotonically (0.31 at $A=4$ to 0.53 at $A=10$); on the frequency axis ($k=3, 4, 5$) and the narrow-Gaussian axis ($\sigma=\pi/5$) the same monotone degradation

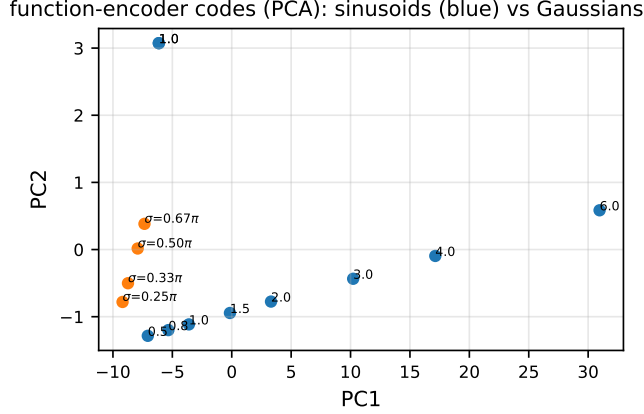


Figure 3: **The encoder represents the specification.** PCA of the learned initial-condition codes: sinusoids (blue) and Gaussians (orange) form distinct, internally ordered clusters (labels: amplitude A and width σ).

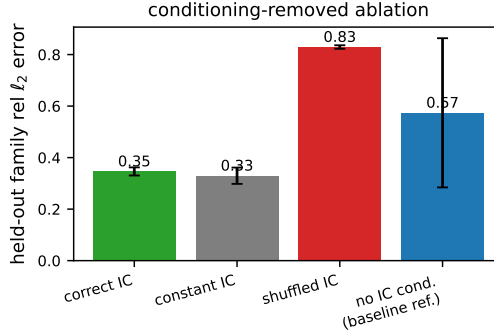


Figure 4: **Conditioning-removed ablation.** Held-out Gaussian-family relative- ℓ_2 error of the structured encoder with the correct initial condition, a constant encoding, a shuffled (wrong) initial condition, and the coefficient-only baseline’s reachable reference. Shuffling degrades past the no-conditioning reference.

216 appears; on the coefficient axis (e.g. $(\nu, \rho) = (6, 2), (2, 6)$) the degradation is mild for reaction-
 217 dominated extrapolation and larger for diffusion- dominated extrapolation. The failure regime is
 218 consistent across seeds and is not a reference artifact (the reaction reference is exact and the reaction-
 219 diffusion reference is converged under refinement, Section 4). The spectral encoding fails *earlier and*
 220 *harder* on the broad-spectrum Gaussian axis than the pointwise representations (Figure 5, center),
 221 indicating that this boundary is a property of the representation rather than of the solution family;
 222 we do not claim a fundamental (capacity-insensitive) boundary because, as noted, a deeper learned
 223 encoder under-fits rather than generalizing differently.

224 **Breadth summary.** A single specification-conditioned model jointly supports **two initial-condition**
 225 **families** (positive sinusoids and Gaussians), **three boundary-condition types** (periodic, Dirichlet,
 226 Neumann), and **three PDE families** (reaction–diffusion, reaction, and diffusion), evaluated on unseen
 227 initial-condition parameters, an unseen initial-condition family, unseen coefficient–initial-condition
 228 combinations, and a held-out boundary condition. The pure reaction family is reported for breadth;
 229 its pointwise logistic map produces sharper solution features, so its absolute errors are higher than
 230 reaction–diffusion (e.g. structured encoder seen 0.1896 versus 0.0724 for the baseline at the fixed
 231 condition), but the same representation ordering and the same queryability result hold.

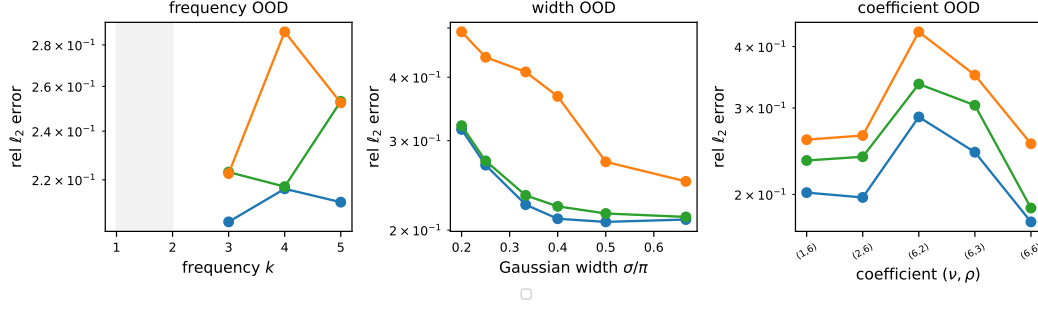


Figure 5: **Out-of-distribution sweeps.** Per-specification relative- ℓ_2 error as the frequency k , the Gaussian width σ , and the coefficients (ν, ρ) move beyond the training distribution, for the raw-input control, the structured encoder, and the spectral basis. Error grows smoothly with novelty; the spectral basis separates from the pointwise representations on the broad-spectrum (narrow Gaussian) axis.

8 Limitations and conclusion

We extended the parameter encoder of a parametric PINN into a problem-specification encoder that conditions on the coefficients, the initial condition, and the boundary condition, and showed that a single model is then queryable at unseen initial-condition amplitudes, an unseen initial-condition function family, unseen coefficient–initial-condition combinations, and a held-out boundary condition, on a non-linear (Fisher) family, with a conditioning-removed ablation and a latent diagnostic confirming genuine, novelty-controlled generalization and no seen-specification regression. The boundary-condition axis is measured against an authored and validated non-periodic reference solver.

Three honest limitations bound this work. First, we did not find a structured initial-condition encoding that beats raw point samples by the pre-registered margin on these smooth one-dimensional initial conditions: the spectral basis, despite aligning with the periodic spectral solvers, degrades on broad-spectrum initial conditions, and a deep learned encoder under-fits. This is itself the central representation finding—the encoding must be aligned with the decoder’s coordinate consumption—and it motivates designing decoders whose coordinate injection matches the representation (e.g., spectral synthesis) as the next step. Second, the boundary-condition axis uses the diffusion family for a clean demonstration; because the bump initial conditions are even, the periodic and homogeneous-Neumann diffusion operators coincide for these data, so the held-out-Neumann result should be read as conditioning on (and generalizing to) an unseen boundary *type label* whose reference coincides with the periodic reference for this family. Third, the diffusion boundary-condition axis exhibits higher absolute error than the reaction–diffusion core, reflecting the known difficulty of enforcing a mass-conserving linear evolution with a soft residual. Extending the boundary-condition conditioning to the reaction–diffusion family with compatible non-even initial conditions, and closing the remaining gap between structured and raw-input conditioning on the initial-condition axis, are the natural next steps.

A Supplementary results

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Table 3: **Per-coefficient non-regression** at the shared condition (fixed initial condition $1 + \sin(x)$, periodic boundaries), relative ℓ_2 error per coefficient, mean over 3 seeds. The structured encoder improves on the baseline at every coefficient (max relative regression 0.0000).

(ν, ρ)	baseline	raw-input control	structured encoder
(1,1)	0.083	0.071	0.065
(1,2)	0.076	0.069	0.063
(1,3)	0.064	0.054	0.050
(1,4)	0.054	0.041	0.041
(1,5)	0.043	0.030	0.037
(2,1)	0.137	0.144	0.132
(2,2)	0.120	0.127	0.115
(2,3)	0.106	0.110	0.099
(2,4)	0.095	0.093	0.084
(2,5)	0.083	0.072	0.073
(3,1)	0.167	0.183	0.162
(3,2)	0.149	0.160	0.144
(3,3)	0.132	0.144	0.127
(3,4)	0.120	0.130	0.108
(3,5)	0.106	0.107	0.094
(4,1)	0.188	0.205	0.180
(4,2)	0.169	0.181	0.163
(4,3)	0.152	0.166	0.145
(4,4)	0.138	0.150	0.125
(4,5)	0.124	0.130	0.110
(5,1)	0.204	0.216	0.192
(5,2)	0.184	0.194	0.177
(5,3)	0.166	0.181	0.159
(5,4)	0.152	0.164	0.141
(5,5)	0.138	0.146	0.123

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